# Git Branching

# Agenda

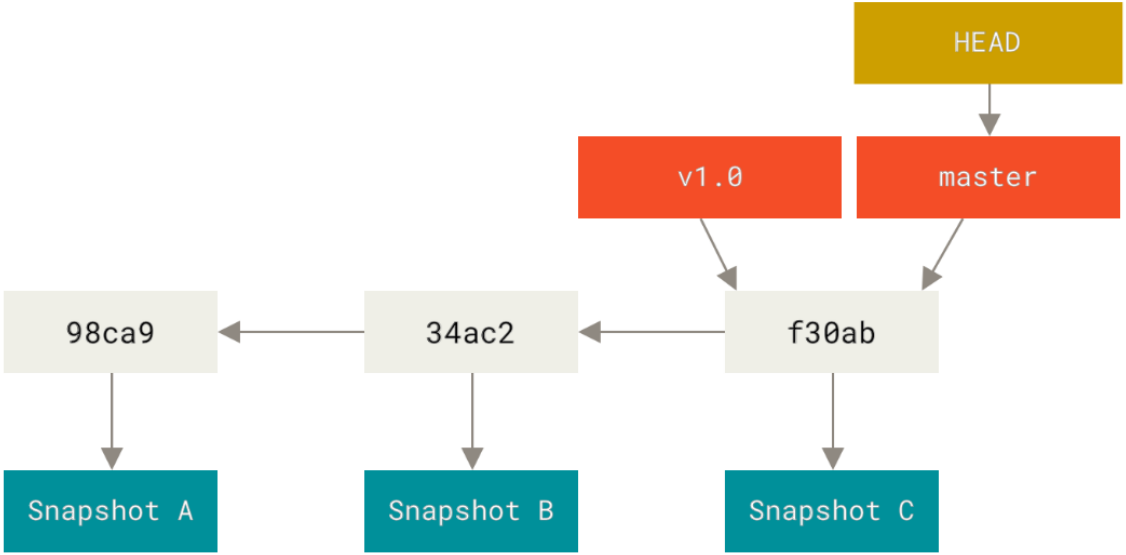
- What is a Branch?
- Creating a Branch
- HEAD Pointer
- Switching Branches
- Divergent History
- Branching and Merging Workflow Example
- Fast-Forward Merge
- Three-Way Merge

**Reference:**

Pro Git: https://github.com/progit/progit2/releases/download/2.1.449/progit.pdf

# What is a Branch?

A branch in Git is simply a lightweight movable pointer to one of these commits. The default branch name in Git is `master`. As you start making commits, you're given a `master` branch that points to the last commit you made. Every time you commit, the `master` branch pointer moves forward automatically.

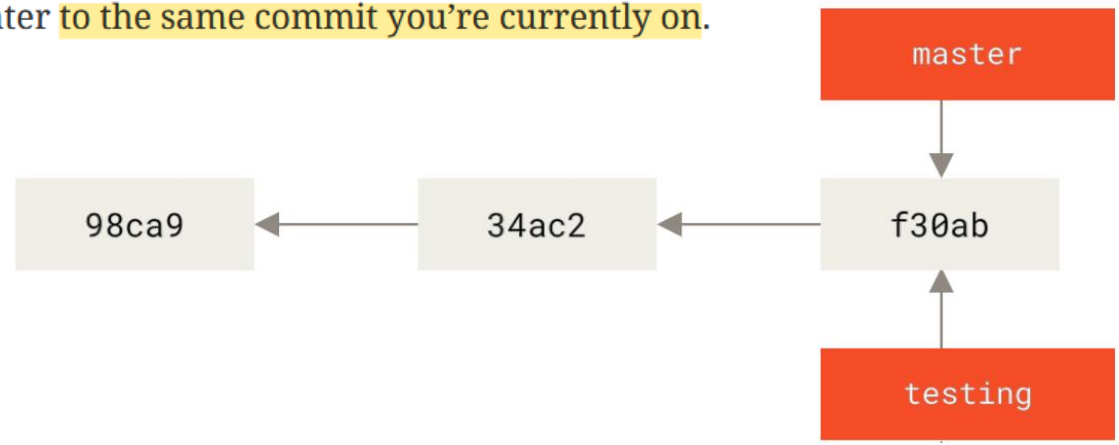


# Creating a Branch

What happens when you create a new branch? Well, doing so creates a new pointer for you to move around. Let's say you want to create a new branch called `testing`. You do this with the `git branch` command:

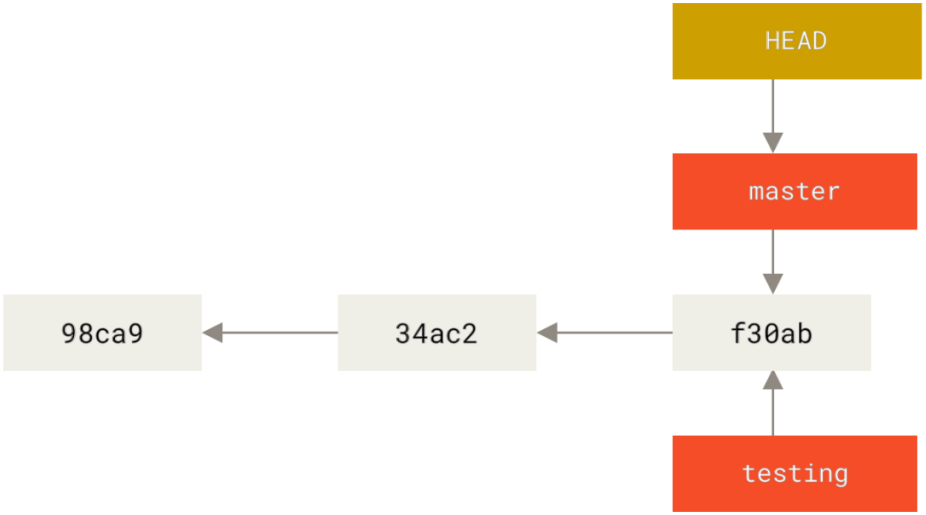
```
$ git branch testing
```

This creates a new pointer to the same commit you're currently on.



# HEAD Pointer

How does Git know what branch you're currently on? It keeps a special pointer called `HEAD`. Note that this is a lot different than the concept of `HEAD` in other VCSs you may be used to, such as Subversion or CVS. In Git, this is a pointer to the local branch you're currently on. In this case, you're still on `master`. The `git branch` command only *created* a new branch — it didn't switch to that branch.



# Git Log --decorate

You can easily see this by running a simple `git log` command that shows you where the branch pointers are pointing. This option is called `--decorate`.

```
$ git log --oneline --decorate
f30ab (HEAD -> master, testing) Add feature #32 - ability to add new formats to the
central interface
34ac2 Fix bug #1328 - stack overflow under certain conditions
98ca9 Initial commit
```

You can see the `master` and `testing` branches that are right there next to the `f30ab` commit.

# Switching Branches

To switch to an existing branch, you run the `git checkout` command. Let's switch to the new `testing` branch:

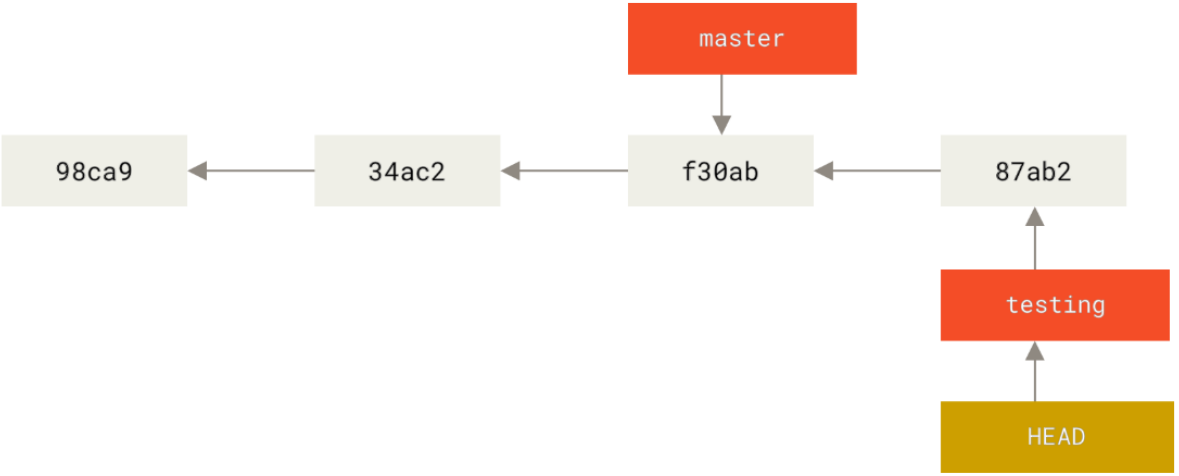
```
$ git checkout testing
```

This moves `HEAD` to point to the `testing` branch.

# Switching Branches

What is the significance of that? Well, let's do another commit:

```
$ vim test.rb
$ git commit -a -m 'Make a change'
```

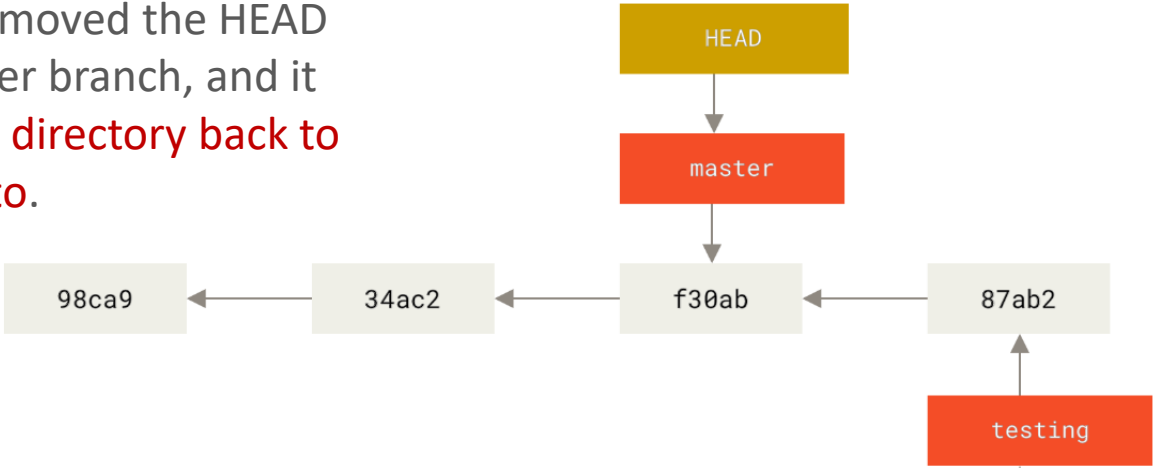


# Divergent History

This is interesting, because now your `testing` branch has moved forward, but your `master` branch still points to the commit you were on when you ran `git checkout` to switch branches. Let's switch back to the `master` branch:

```
$ git checkout master
```

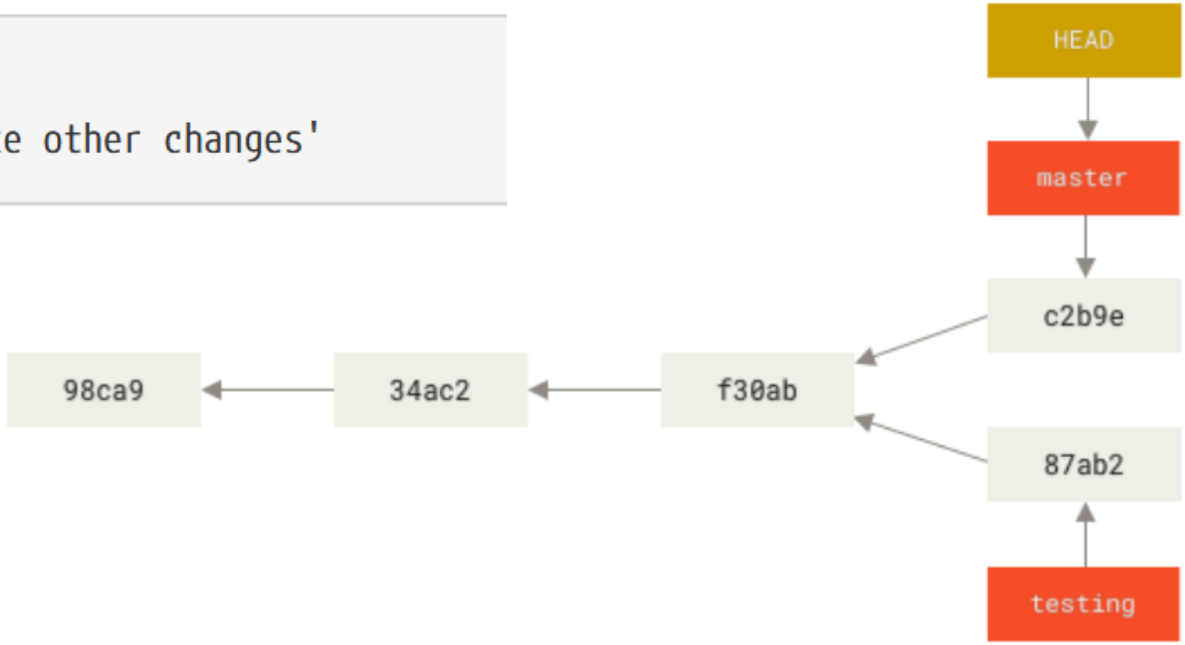
That command did two things. It moved the HEAD pointer back to point to the master branch, and it reverted the files in your working directory back to the snapshot that master points to.



# Divergent History

Let's make a few changes and commit again:

```
$ vim test.rb
$ git commit -a -m 'Make other changes'
```



# Divergent History

NOTE: git log will only show commit history below the branch you've checked out.

To show commit history for the desired branch you have to explicitly specify it: `git log testing`. To show all of the branches, add `--all` to your `git log` command.

You can also see this easily with the `git log` command. If you run `git log --oneline --decorate --graph --all` it will print out the history of your commits, showing where your branch pointers are and how your history has diverged.

```
$ git log --oneline --decorate --graph --all
* c2b9e (HEAD, master) Make other changes
| * 87ab2 (testing) Make a change
|/
* f30ab Add feature #32 - ability to add new formats to the central interface
* 34ac2 Fix bug #1328 - stack overflow under certain conditions
* 98ca9 Initial commit of my project
```

# More on Switching

*Creating a new branch and switching to it at the same time*

It's typical to create a new branch and want to switch to that new branch at the same time — this can be done in one operation with `git checkout -b <newbranchname>`.

From Git version 2.23 onwards you can use `git switch` instead of `git checkout` to:

- Switch to an existing branch: `git switch testing-branch`.
- Create a new branch and switch to it: `git switch -c new-branch`. The `-c` flag stands for create, you can also use the full flag: `--create`.
- Return to your previously checked out branch: `git switch -`.

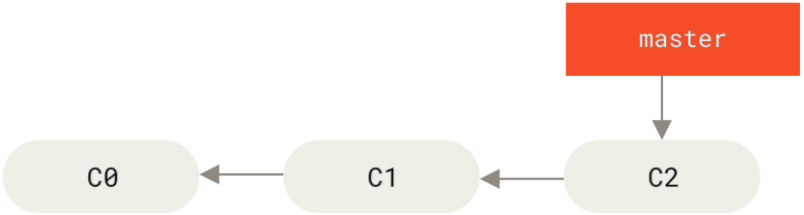
# Branching and Merging Workflow

Let's go through a simple example of branching and merging with a workflow that you might use in the real world. You'll follow these steps:

1. Do some work on a website.
2. Create a branch for a new user story you're working on.
3. Do some work in that branch.

At this stage, you'll receive a call that another issue is critical and you need a hotfix. You'll do the following:

1. Switch to your production branch.
2. Create a branch to add the hotfix.
3. After it's tested, merge the hotfix branch, and push to production.
4. Switch back to your original user story and continue working.



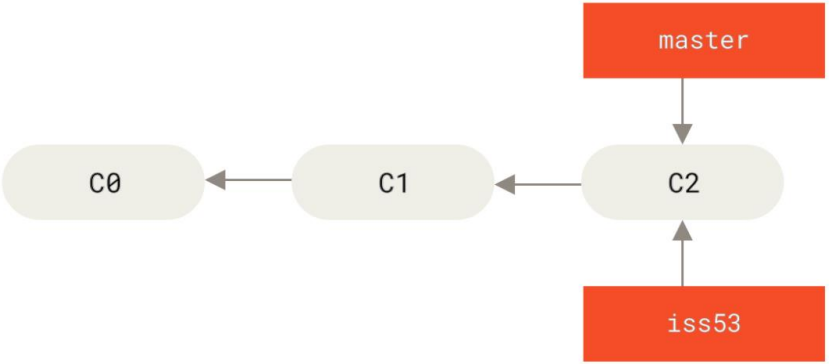
# Workflow: Creating branch for issue #53

You've decided that you're going to work on issue #53 in whatever issue-tracking system your company uses. To create a new branch and switch to it at the same time, you can run the `git checkout` command with the `-b` switch:

```
$ git checkout -b iss53
Switched to a new branch "iss53"
```

This is shorthand for:

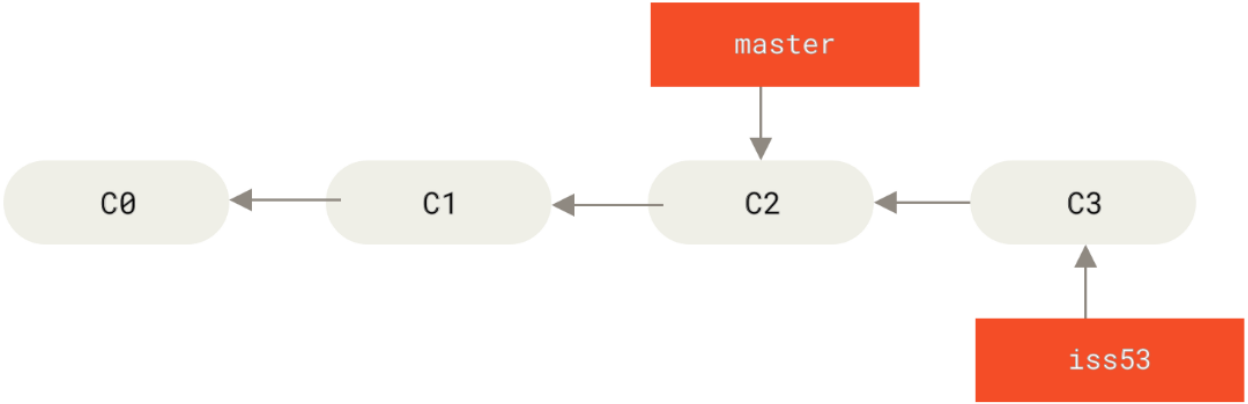
```
$ git branch iss53
$ git checkout iss53
```



# Workflow: Do some work

You work on your website and do some commits. Doing so moves the iss53 branch forward, because you have it checked out (that is, your HEAD is pointing to it):

```
$ vim index.html
$ git commit -a -m 'Create new footer [issue 53]'
```

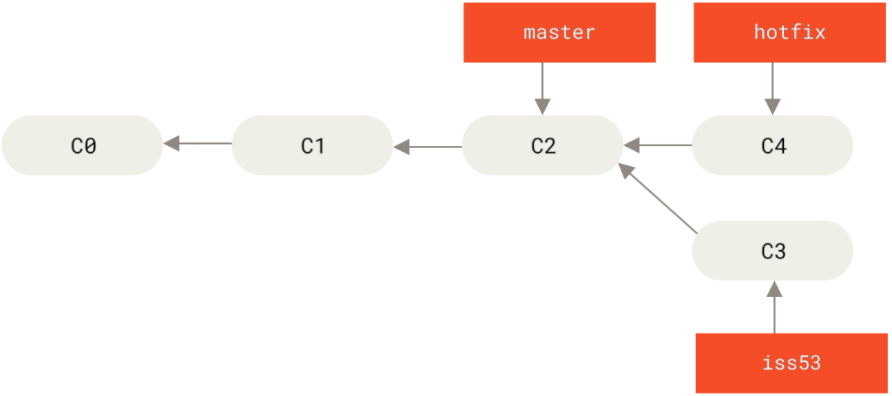


# Workflow: Create hotfix branch & do some work

```
$ git checkout master
Switched to branch 'master'
```

Next, you have a hotfix to make. Let's create a `hotfix` branch on which to work until it's completed:

```
$ git checkout -b hotfix
Switched to a new branch 'hotfix'
$ vim index.html
$ git commit -a -m 'Fix broken email address'
[hotfix 1fb7853] Fix broken email address
 1 file changed, 2 insertions(+)
```



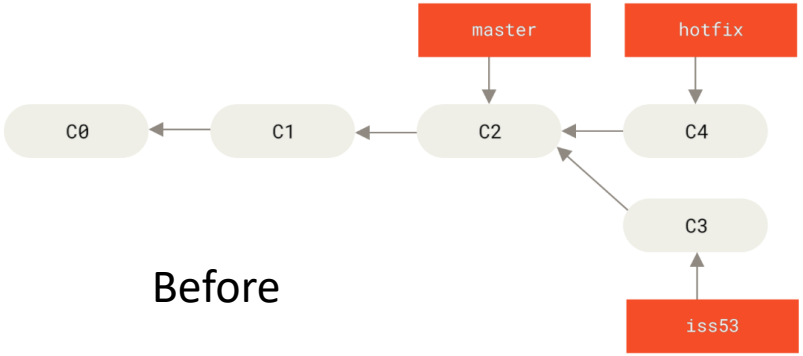
# Workflow: Merge master and hotfix

## Fast-Forward Merge

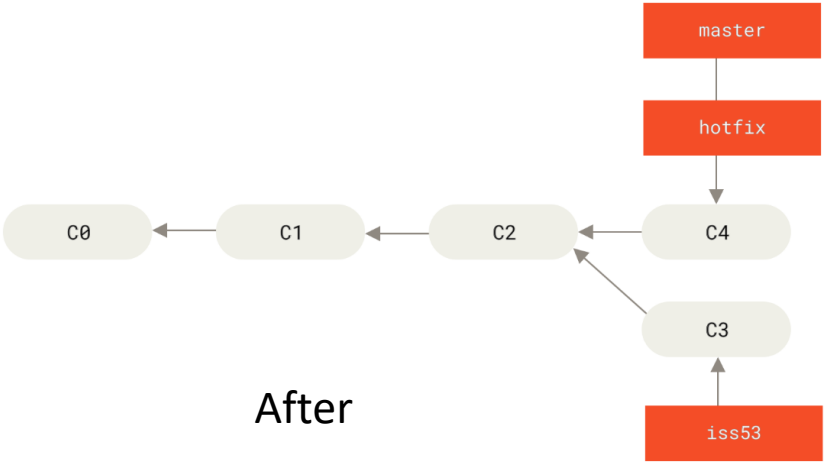
You can run your tests, make sure the hotfix is what you want, and finally merge the `hotfix` branch back into your `master` branch to deploy to production. You do this with the `git merge` command:

```
$ git checkout master
$ git merge hotfix
Updating f42c576..3a0874c
Fast-forward
 index.html | 2 ++
 1 file changed, 2 insertions(+)
```

`master` *is fast-forwarded to* `hotfix`




Before




After

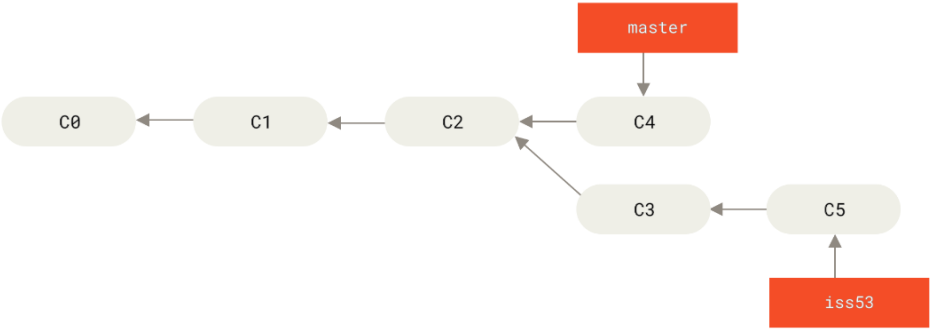
# Workflow: Delete hotfix branch & switch back to issue#53 branch

After your super-important fix is deployed, you're ready to switch back to the work you were doing before you were interrupted. However, first you'll delete the `hotfix` branch, because you no longer need it — the `master` branch points at the same place. You can delete it with the `-d` option to `git branch`:

```
$ git branch -d hotfix
Deleted branch hotfix (3a0874c).
```

Now you can switch back to your work-in-progress branch on issue #53 and continue working on it.

```
$ git checkout iss53
Switched to branch "iss53"
$ vim index.html
$ git commit -a -m 'Finish the new footer [issue 53]'
[iss53 ad82d7a] Finish the new footer [issue 53]
1 file changed, 1 insertion(+)
```

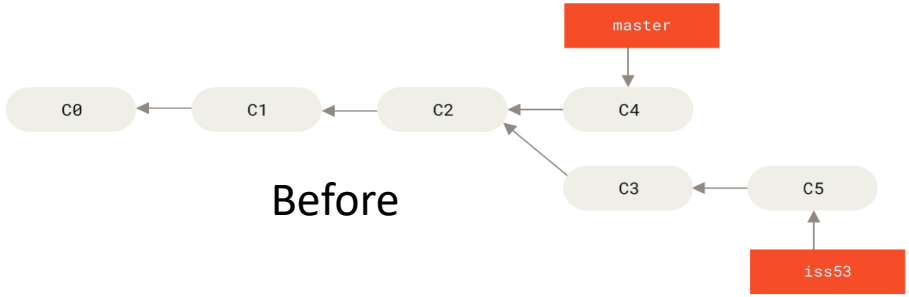


# Workflow: Merging issue#53 work into master branch

## Three-Way Merge

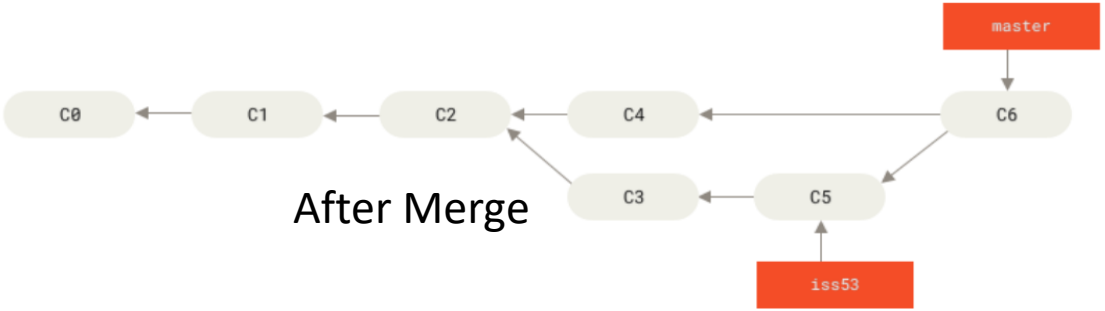
Suppose you've decided that your issue #53 work is complete and ready to be merged into your `master` branch. In order to do that, you'll merge your `iss53` branch into `master`, much like you merged your `hotfix` branch earlier. All you have to do is check out the branch you wish to merge into and then run the `git merge` command:

```
$ git checkout master
Switched to branch 'master'
$ git merge iss53
Merge made by the 'recursive' strategy.
index.html |    1 +
1 file changed, 1 insertion(+)
```




Before

```
$ git branch -d iss53
```




After Merge

# More on Branching Commands

The `git branch` command does more than just create and delete branches. If you run it with no arguments, you get a simple listing of your current branches:

```
$ git branch
  iss53
* master
  testing
```

The useful `--merged` and `--no-merged` options can filter this list to branches that you have or have not yet merged into the branch you're currently on. To see which branches are already merged into the branch you're on, you can run `git branch --merged`:

```
$ git branch --merged
  iss53
* master
```

Because you already merged in `iss53` earlier, you see it in your list. Branches on this list without the `*` in front of them are generally fine to delete with `git branch -d`; you've already incorporated their work into another branch, so you're not going to lose anything.

# More on Branching Commands

To see all the branches that contain work you haven't yet merged in, you can run `git branch --no-merged`:

```
$ git branch --no-merged
  testing
```

This shows your other branch. Because it contains work that isn't merged in yet, trying to delete it with `git branch -d` will fail:

```
$ git branch -d testing
error: The branch 'testing' is not fully merged.
If you are sure you want to delete it, run 'git branch -D testing'.
```

If you really do want to delete the branch and lose that work, you can force it with `-D`, as the helpful message points out.

# More on Branching Commands

You can always provide an additional argument to ask about the merge state with respect to some other branch without checking that other branch out first, as in, what is not merged into the `master` branch?

```
$ git checkout testing
$ git branch --no-merged master
  topicA
  featureB
```